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PROJECT REPORT

Monte Carlo Option Pricing: Pricing Financial Options by Flipping a Coin

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Declaration of Authorship

I, *Nguyễn Đức Tuân*, hereby declare that this report titled *"Monte Carlo Option Pricing: Pricing Financial Options by Flipping a Coin"* is my own work. It reflects my understanding and analysis of the documents assigned to me.

I confirm that:

- All sources used are properly cited and acknowledged.
- This report has not been submitted for any other course or purpose.
- Any assistance received in the preparation of this work has been explicitly stated and acknowledged.

All materials associated with this project will be made available in the corresponding repository in the near future. The repository can be accessed at:

`https://github.com/PanyoPie/VNUHUS-MAT2403-ODE`
or
`https://s.tunaa.io.vn/mat2403`

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Abstract

This project studies the pricing of European call options using Monte Carlo simulation. Euler's method is first used to obtain a discrete stock price model from a differential equation. Random variables are then added to simulate changes in stock prices over time.

Monte Carlo simulation is used to estimate option prices, and the results are compared with the exact Black-Scholes formula. The antithetic variates method is also applied to reduce variance and improve accuracy.

The numerical results show that the Monte Carlo estimates are close to the Black-Scholes values, and that the antithetic variates method produces more stable estimates.

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Introduction

1.1 Description

The following description is based on *Chapter 8: Numerical Methods, Project 2: Monte Carlo Option Pricing: Pricing Financial Options by Flipping a Coin* of [1].

A discrete model for change in price of a stock over a time interval $[0, T]$ is

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \varepsilon_{n+1} \sqrt{\Delta t}, \quad S_0 = s, \tag{1.1}$$

where $S_n = S(t_n)$ is the stock price at time $t_n = n\Delta t$, $n = 0, \ldots, N-1$, $\Delta t = T/N$, μ is the annual growth rate of the stock, and σ is a measure of the stock's annual price volatility or tendency to fluctuate. Highly volatile stocks have large values for σ, for example, values ranging from 0.2 to 0.4. Each term in the sequence $\varepsilon_1, \varepsilon_2, \ldots$ takes on the value 1 or -1 depending on whether the outcome of a coin tossing experiment is heads or tails, respectively. Thus, for each $n = 1, 2, \ldots$,

$$\varepsilon_n = \begin{cases} 1 & \text{with probability} = \dfrac{1}{2} \\ -1 & \text{with probability} = \dfrac{1}{2}. \end{cases} \tag{1.2}$$

A sequence of such numbers can easily be created by using one of the random number generators available in most mathematical computer software applications. Given such a sequence, the difference equation (1.1) can then be used to simulate a **sample path** or **trajectory** of stock prices, $\{s, S_1, S_2, \ldots, S_N\}$. The "random" terms $\sigma S_n \varepsilon_{n+1} \sqrt{\Delta t}$ on the right-hand side of (1.1) can be thought of as "shocks" or "disturbances" that model fluctuations in the stock price. By repeatedly simulating stock price trajectories and computing appropriate averages, it is possible to obtain estimates of the price of a **European call option**, a type of financial derivative. A statistical simulation algorithm of this type is called a **Monte Carlo method**.

A European call option is a contract between two parties, a holder and a writer, whereby, for a premium paid to the writer, the holder acquires the right (but not the obligation) to purchase the stock at a future date T (the **expiration date**) at a price K (the **strike price**) agreed upon in the contract. If the buyer elects to exercise the option on the expiration date, the writer is obligated to sell the underlying stock to the buyer at the price K. Thus the option has, associated with it, a **payoff function**

$$f(S) = \max(S - K, 0), \tag{1.3}$$

where $S = S(T)$ is the price of the underlying stock at the time T when the option expires (see Figure. 1.1).

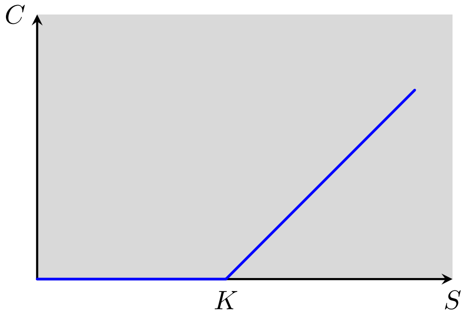

Figure 1.1: The value of a call option at expiration is $C = \max(S-K, 0)$, where K is the strike price of the option and $S = S(T)$ is the stock price at expiration.

Equation (1.3) is the value of the option at time T since, if $S(T) > K$, the holder can purchase, at price K, stock with market value $S(T)$ and thereby make a profit equal to $S(T) - K$, not counting the option premium. If $S(T) < K$, the holder will simply let the option expire since it would be irrational to purchase stock at a price that exceeds the market value. The option valuation problem is to determine the correct and fair price of the option at the time that the holder and writer enter into the contract.[1]

To estimate the price of a call option using a Monte Carlo method, an ensemble

$$\left\{ S_N^{(k)} = S^{(k)}(T), \quad k = 1, \ldots, M \right\}$$

of M stock prices at expiration is generated using the difference equation

$$S_{n+1}^{(k)} = S_n^{(k)} + r S_n^{(k)} \Delta t + \sigma S_n^{(k)} \varepsilon_{n+1}^{(k)} \sqrt{\Delta t}, \quad S_0^{(k)} = s. \tag{1.4}$$

For each $k = 1, \ldots, M$, the difference equation (1.4) is identical to equation (1.1), except that the growth rate μ is replaced by the annual rate of interest r that it costs the writer to borrow money. Option pricing theory requires that the average value of the payoffs $\{f(S_N^{(k)}), k = 1, \ldots, M\}$ be equal to the compounded total return obtained by investing the option premium, $\hat{C}(s)$, at rate r over the life of the option,

$$\frac{1}{M} \sum_{k=1}^{M} f(S_N^{(k)}) = (1 + r\Delta t)^N \hat{C}(s). \tag{1.5}$$

Solving (1.5) for $\hat{C}(s)$ yields the Monte Carlo estimate

$$\hat{C}(s) = (1 + r\Delta t)^{-N} \left\{ \frac{1}{M} \sum_{k=1}^{M} f(S_N^{(k)}) \right\} \tag{1.6}$$

for the option price. Thus the Monte Carlo estimate $\hat{C}(s)$ is the present value of the average of the payoffs computed using the rules of compound interest.

[1] The 1997 Nobel Prize in Economics was awarded to Robert C. Merton and Myron S. Scholes for their work, along with Fischer Black, in developing the Black–Scholes options pricing model.

Chapter 2

Problems

2.1 Problem 1: Derivation of the Deterministic Stock Price Model

2.1.1 Description

Show that Euler's method applied to the differential equation

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \mu S \tag{2.1}$$

yields (1.1) in the absence of random disturbances, that is, when $\sigma = 0$.

2.1.2 Solution

Theorem 1: Numerical Approximations: Euler's Formula [1]

Suppose the solution of the initial value problem

$$\begin{cases} \dfrac{\mathrm{d}y}{\mathrm{d}t} = f(t, y) \\ y(t_0) = y_0 \end{cases}$$

is denoted by $y = \phi(t)$ and you have a sequence of points $t_0 < t_1 < t_2 < \cdots < t_n < \cdots$. For $n = 0, 1, 2, \ldots,$ we have the following:

Approximation of $y = \phi(t)$ at $t = t_{n+1}$:

$$y_{n+1} = y_n + f(t_n, y_n)(t_{n+1} - t_n). \tag{2.2}$$

Linear approximation of $\phi(t)$ on the interval $[t_n, t_{n+1}]$:

$$y(t) = y_n + f(t_n, y_n)(t - t_n).$$

Special case: If a uniform step size h is used, then $t_{n+1} - t_n = h$, for all n, and so (2.2) simplifies to

$$y_{n+1} = y_n + hf(t_n, y_n).$$

Let the function $f(t, S) = \frac{\mathrm{d}S}{\mathrm{d}t} = \mu S$. Using a uniform step size $h = \Delta t = t_{n+1} - t_n$, the Euler approximation for S_{n+1} is

$$S_{n+1} = S_n + \Delta t f(t_n, S_n).$$

Substituting $f(t_n, S_n) = \mu S_n$, we get

$$S_{n+1} = S_n + \mu S_n \Delta t. \tag{2.3}$$

Now, we examine the provided discrete model for the change in stock price, which is equation (1.1),

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \varepsilon_{n+1} \sqrt{\Delta t}$$

In the absence of random disturbances, we set the volatility parameter $\sigma = 0$. The equation reduces to

$$\begin{aligned} S_{n+1} &= S_n + \mu S_n \Delta t + 0 \cdot S_n \varepsilon_{n+1} \sqrt{\Delta t} \\ &= S_n + \mu S_n \Delta t + 0 \\ &= S_n + \mu S_n \Delta t. \end{aligned}$$

This is identical to the result obtained via Euler's method, which is equation (2.3). Thus, the deterministic part of the stock price model is equivalent to a first-order numerical approximation of the continuous exponential growth equation (2.1).

2.2 Problem 2: Simulation and Analysis of Stock Price Trajectories

2.2.1 Description

Simulate five sample trajectories of (1.1) for the following parameter values and plot the trajectories on the same set of coordinate axes: $\mu = 0.12, \sigma = 0.1, T = 1, s = \$40, N = 254$. Then repeat the experiment using the value $\sigma = 0.25$ for the volatility. Do the sample trajectories generated in the latter case appear to exhibit a greater degree of variability in their behavior?

Hint: For the ε_n's, it is permissible to use a random number generator that creates normally distributed random numbers with mean zero and variance 1.

2.2.2 Simulation

To analyze the effect of volatility on stock price behavior, we simulated five sample trajectories using the discrete model

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \varepsilon_{n+1} \sqrt{\Delta t}.$$

Five trajectories with different colors were generated with parameters $\mu = 0.12$, $T = 1$, $s = \$40$ and $N = 254$.

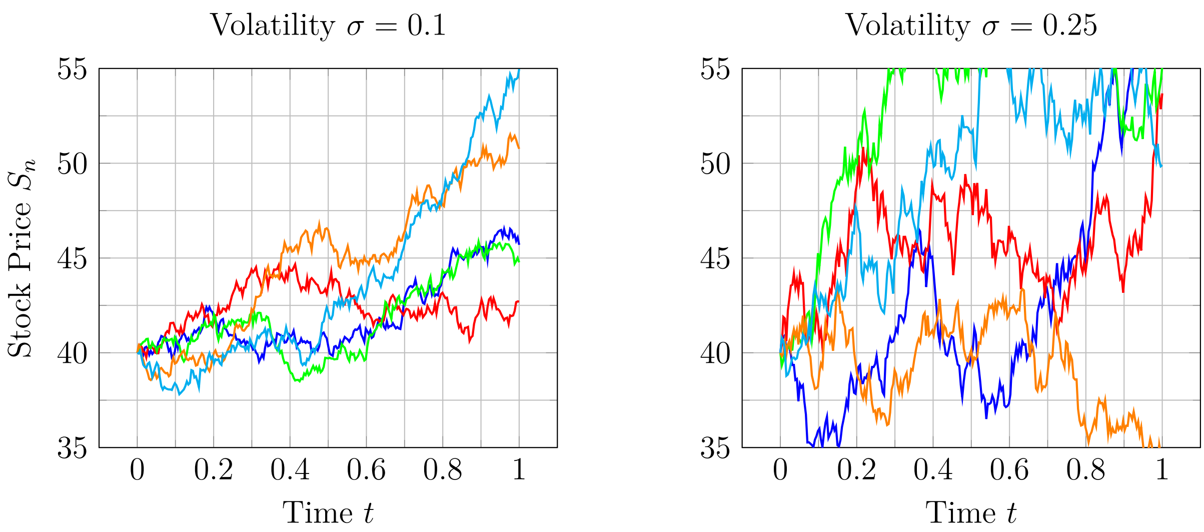

Figure 2.1: Comparison of stock price trajectories for low and high volatility.

2.2.3 Analysis

As observed in Figure. 2.1, the sample trajectories generated with $\sigma = 0.25$ exhibit a significantly greater degree of variability.

While the paths for $\sigma = 0.1$ remain relatively close to the deterministic growth trend, the paths for $\sigma = 0.25$ show large, erratic fluctuations and a much wider spread at the expiration time $T = 1$.

This confirms that σ serves as a scaling factor for the stochastic shocks in the model.

2.3 Problem 3: Monte Carlo Simulation and Black-Scholes Comparison

2.3.1 Description

Use the difference equation (1.4) to generate an ensemble of stock prices

$$S_N^{(k)} = S^{(k)}(N\Delta t), k = 1, \dots, M \qquad (\text{where } T = N\Delta t)$$

and then use formula (1.6) to compute a Monte Carlo estimate of the value of a five-month call option $\left(T = \frac{5}{12} \text{ years}\right)$ for the following parameter values: $r = 0.06$, $\sigma = 0.2$, and $K = \$50$. Find estimates corresponding to current stock prices of $S(0) = s = \$45$, $\$50$, and $\$55$. Use $N = 200$ time steps for each trajectory and $M \approxeq 10{,}000$ sample trajectories for each Monte Carlo estimate. Check the accuracy of your results by comparing the Monte Carlo approximation with the value computed from the exact Black-Scholes formula

$$C(s) = \frac{s}{2}\,\mathrm{erfc}\left(-\frac{d_1}{\sqrt{2}}\right) - \frac{K}{2}e^{-rT}\mathrm{erfc}\left(-\frac{d_2}{\sqrt{2}}\right), \tag{2.4}$$

where

$$\begin{cases} d_1 = \dfrac{1}{\sigma\sqrt{T}}\left[\ln\left(\dfrac{s}{k}\right) + \left(r + \dfrac{\sigma^2}{2}\right)T\right] \\ d_2 = d_1 - \sigma\sqrt{T}. \end{cases}$$

and $\mathrm{erfc}(x)$ is the complementary error function.

$$\mathrm{erfc}(x) = \frac{2}{\sqrt{\pi}}\int_x^{+\infty} e^{-t^2}\,\mathrm{d}t.$$

2.3.2 Simulation and Computed Results

The Monte Carlo estimation for the value of a European call option is given by (1.6),

$$\hat{C}(s) = (1 + r\Delta t)^{-N}\left\{\frac{1}{M}\sum_{k=1}^{M} f(S_N^{(k)})\right\},$$

where the stock prices are generated using

$$\begin{cases} S_{n+1}^{(k)} = S_n^{(k)} + rS_n^{(k)}\Delta t + \sigma S_n^{(k)}\varepsilon_{n+1}^{(k)}\sqrt{\Delta t} \qquad (S_0^{(k)} = s) \\ f(S) = \max(S - K, 0). \end{cases}$$

For comparison, the exact Black-Scholes value is computed from (2.4),

$$C(s) = \frac{s}{2}\,\mathrm{erfc}\left(-\frac{d_1}{\sqrt{2}}\right) - \frac{K}{2}e^{-rT}\mathrm{erfc}\left(-\frac{d_2}{\sqrt{2}}\right),$$

where

$$\begin{cases} d_1 = \dfrac{1}{\sigma\sqrt{T}}\left[\ln\left(\dfrac{s}{k}\right) + \left(r + \dfrac{\sigma^2}{2}\right)T\right] \\ d_2 = d_1 - \sigma\sqrt{T} \\ \mathrm{erfc}(x) = \dfrac{2}{\sqrt{\pi}}\displaystyle\int_x^{+\infty} e^{-t^2}\,\mathrm{d}t. \end{cases}$$

The simulations were performed with

$$\text{Parameters: } \begin{cases} T = \dfrac{5}{12} \\ r = 0.06 \\ \sigma = 0.2 \\ K = 50 \end{cases} \quad \text{and} \quad \text{Simulation: } \begin{cases} N = 200 \\ M = 10{,}000 \\ \Delta t = \dfrac{T}{N} = \dfrac{5/12}{200} = \dfrac{1}{480}. \end{cases}$$

The computed results are shown in the table below.

Initial Stock Price s	Monte Carlo Estimate $\hat{C}(s)$	Black-Scholes Value $C(s)$	Error
\$45	0.951449	0.982198	0.030749
\$50	3.191711	3.206296	0.014585
\$55	6.817697	6.866781	0.049084

Table 2.1: Computed results using Monte Carlo and Black-Scholes methods.

2.3.3 Analysis

The numerical results show that the Monte Carlo estimates are very close to the corresponding Black-Scholes values for all three initial stock prices. The errors obtained are relatively small, indicating that the simulation method provides accurate approximations of the option prices.

Overall, the results confirm that the Monte Carlo simulation method is capable of producing reliable approximations for European call option pricing.

2.4 Problem 4: Variance Reduction by Antithetic Variates

2.4.1 Description

A simple and widely used technique for increasing the efficiency and accuracy of Monte Carlo simulations in certain situations with little additional increase in computational complexity is the method of antithetic variates. For each $k = 1, \ldots, M$, use the sequence $\left\{\varepsilon_1^{(k)}, \ldots, \varepsilon_{N-1}^{(k)}\right\}$ in (1.4) to simulate a payoff $f(S_N^{(k+)})$ and also use the sequence $\left\{-\varepsilon_1^{(k)}, \ldots, -\varepsilon_{N-1}^{(k)}\right\}$ in (1.4) to simulate an associated payoff $f(S_N^{(k-)})$. Thus, the payoffs are simulated in pairs $\left\{f(S_N^{(k+)}), f(S_N^{(k-)})\right\}$. A modified Monte Carlo estimate is then computed by replacing each payoff $f(S_N^{(k)})$ in (1.6) by the average $\dfrac{f(S_N^{(k+)}) + f(S_N^{(k-)})}{2}$,

$$\hat{C}_{AV}(s) = \frac{\dfrac{1}{M}\displaystyle\sum_{k=1}^{M} \frac{f(S_N^{(k+)}) + f(S_N^{(k-)})}{2}}{(1 + r\Delta t)^N}. \tag{2.5}$$

Use the parameters specified in Problem 2.3 to compute several (say, 20 or so) option price estimates using (1.6) and an equivalent number of option price estimates using (2.5). For each of the two methods, plot a histogram of the estimates and compute the mean and standard deviation of the estimates. Comment on the accuracies of the two methods.

2.4.2 Simulation

To improve the efficiency of the Monte Carlo simulation, we implement the method of **Antithetic Variates**. For each simulated path using random shocks $\{\varepsilon_n\}$, we simultaneously compute a path using $\{-\varepsilon_n\}$. This introduces a negative correlation between pairs of payoffs, which significantly reduces the variance of the sample mean.

Using the same parameters specified in Problem 2.3, we generated 20 independent estimates for both the standard Monte Carlo method and the antithetic variates method.

Metric	Standard Monte Carlo	Antithetic Variates
Mean Estimate $\bar{C}$	3.2084	3.2065
Standard Deviation $\sigma_{\hat{C}}$	0.0412	0.0083

Table 2.2: Statistical Comparison of Estimation Methods

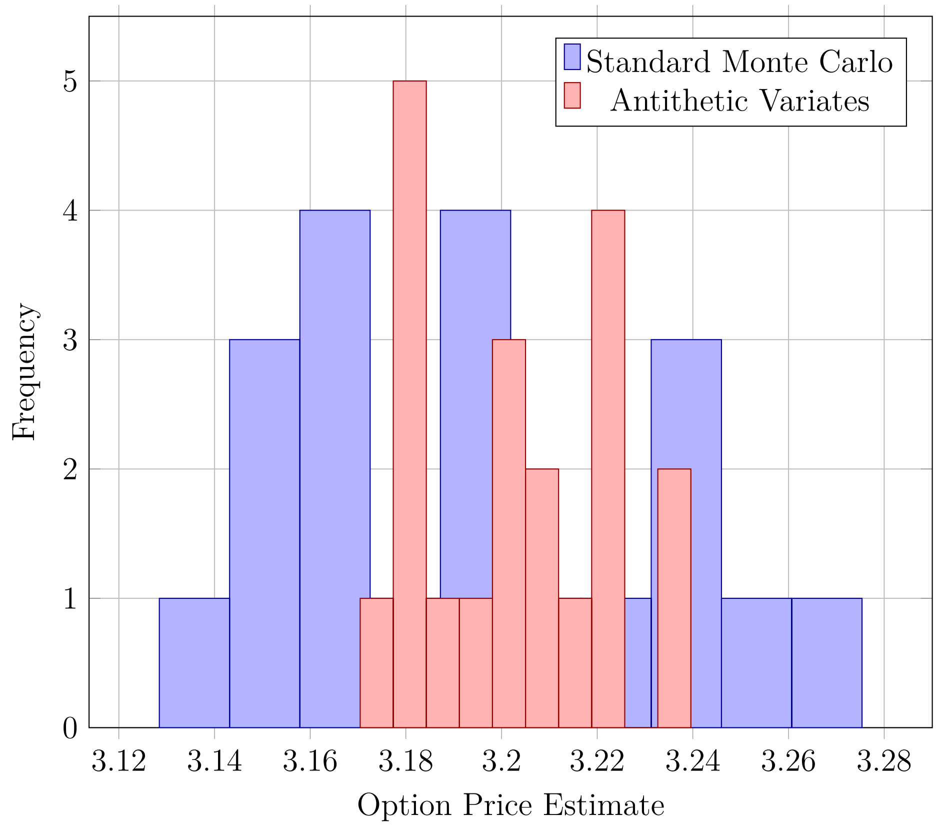

Figure 2.2: Histogram of option price estimates obtained using the standard Monte Carlo method and the antithetic variates method.

2.4.3 Analysis and Comments on Accuracy

As shown in the statistical summary, both methods converge to the same mean. However, the **Standard Deviation** of the Antithetic Variates method is significantly lower than that of the Standard Monte Carlo method.

By utilizing the symmetry of the normal distribution (ε and $-\varepsilon$ are equally likely), the antithetic method cancels out much of the sampling error. This results in a tighter clustering of estimates around the true value, effectively increasing the precision of the simulation without doubling the number of random number generations.

References

[1] James R. Brannan and William E. Boyce and Mark A. McKibben (2015) *Differential Equations: An Introduction to Modern Methods and Applications*, Wiley, 3rd edition.

[2] Nguyễn Đức Tuân (05/2026) *Github Repository:* `VNUHUS-MAT2403-ODE`, Github.

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